service include claims for principal ID, role, and device trust (from the embedded posture client). When the principal is a customer, extra data—such as account status or fraud score—is fetched from ACME's customer database during login and stored for the duration of the token.

- For employees using the same platform, the PDP enhances token data with HR attributes (such as employment status and department) and Finance signals (such as authorization limits). The PIP gathers this information from internal APIs during login or periodically through background synchronization.
- The engineering deployment tool uses a hybrid approach. The token maintains employee identity and device trust, while the PDP queries PagerDuty in real time to verify that the engineer is currently on call. This reduces the risk of outdated access decisions while keeping most data local.

Each of these options represents a point in the three-dimensional space introduced earlier:

- Which signals are required?
- How are they delivered?
- How fresh should they be?

The next section examines the third dimension in detail—freshness, caching, and revocation—and how systems like ACME balance latency with assurance.

12.4 Freshness, caching, and revocation

The third dimension in the authorization context choice space is freshness: How current a signal needs to be when a policy decision is made. Authorization decisions are only as reliable as the data they rely on. If a signal becomes stale, policies might grant access long after the underlying facts have changed.

Every organization must balance latency, load, and assurance when deciding how fresh its signals need to be. The ideal is instantaneous accuracy, but real systems must manage performance and availability constraints.